

Learning Latent Representations for Neural Surrogate Models of PDEs

From Proper Orthogonal Decomposition to Frequency-Conditioned Autoencoders

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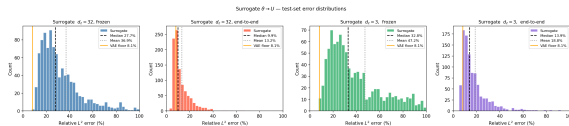
Introduction: Preliminary Stationary Case

Goal: Emulate parameter-to-field map $\theta \mapsto U$ via a compact latent representation.

Method: Train a VAE ($U \rightarrow z$) followed by a latent MLP surrogate ($\theta \rightarrow z$).

Key findings:

- A frozen decoder yields very poor performance (27.7% error).
- **End-to-End Fine-tuning** is indispensable (reduces error to 9.9%, almost reaching the VAE reconstruction floor of 8.1%).

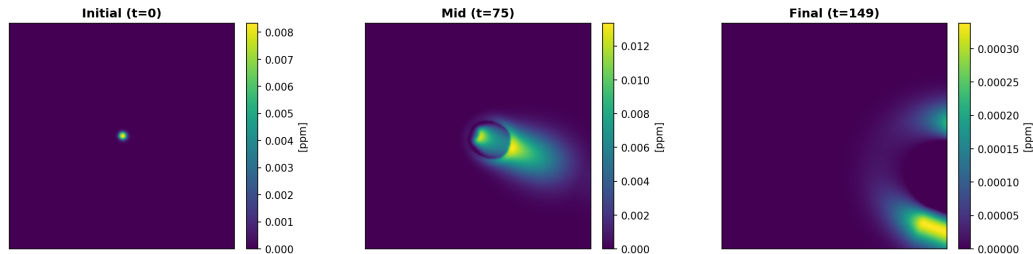


The Transient Problem

Dataset: Atmospheric Methane (CH_4) Dispersion

- 8,100 simulations, 150 snapshots on 200×200 grid.
- Parameters: $\theta = (k, A, C)$ (reaction rate, angle, concentration).

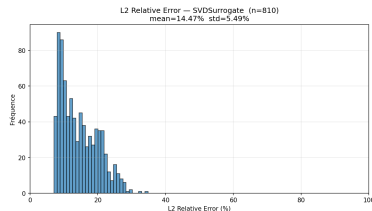
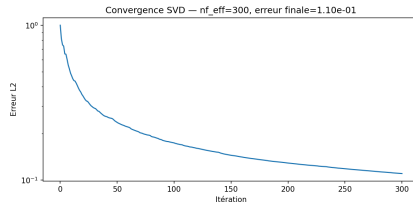
CH4 Transient Field (Simulation #7270, $k=0.936$, $A=340.000$, $C=0.011$)



Challenge: High dimensionality (150×200^2) makes direct regression intractable.

Linear Baseline 1: Tucker POD

- **Method:** Tucker decomposition into separable (F, P, G) modes.
- **Memory Bottleneck:** Requires holding the full tensor in RAM. Forces severe spatial downsampling from 200×200 to 40×40 .
- **Performance:** Reaches an irreducible $\sim 10\%$ reconstruction floor.

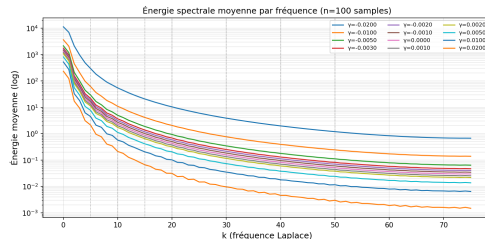


Temporal Decoupling: The Laplace Domain

Transform temporal fields via **discrete Bromwich integral**:

$$\hat{U}(s_k) = \int_0^T U(t) e^{-s_k t} dt \quad (\text{differentiable})$$

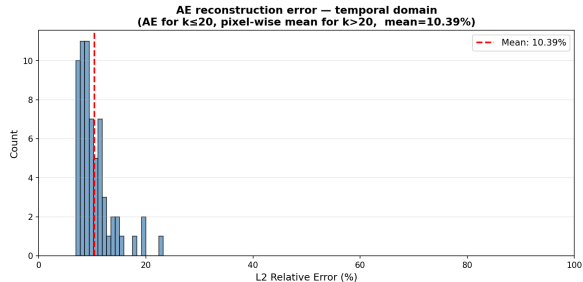
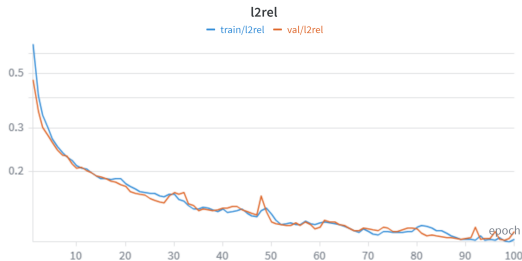
- Converts 3D transient fields into 2D independent stationary-like complex fields.
- Spectrum drops rapidly: truncated at $k_{\max} = 20$.
- Allows processing the problem at the **full original spatial resolution**.



Latent Space: Shared Architecture and Temporal Error

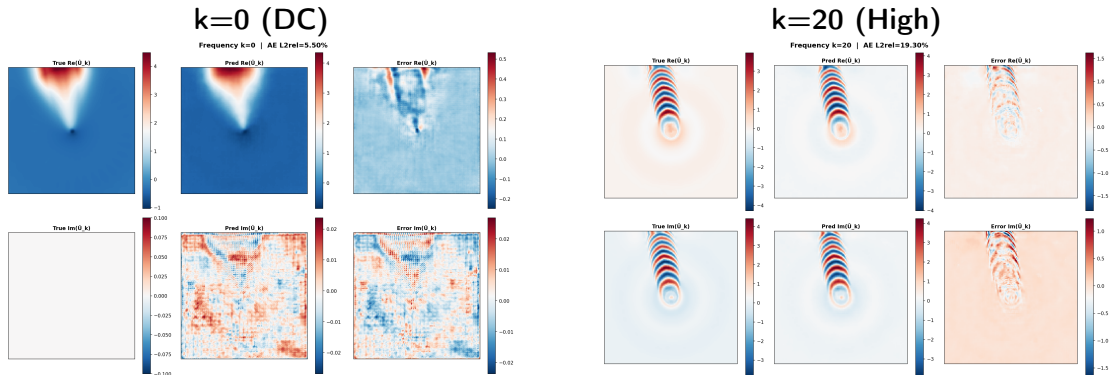
We train a **single shared autoencoder** using **FiLM modulation** and **Sinusoidal Encodings**.

Temporal inversion yields a **9.94% mean error floor**.



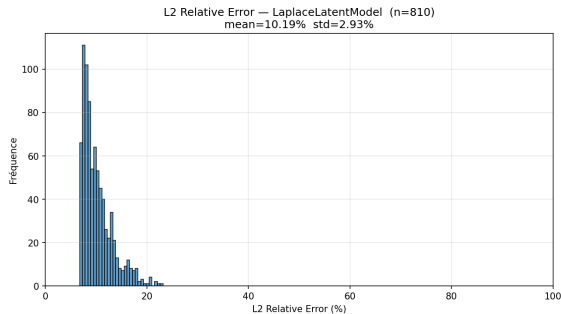
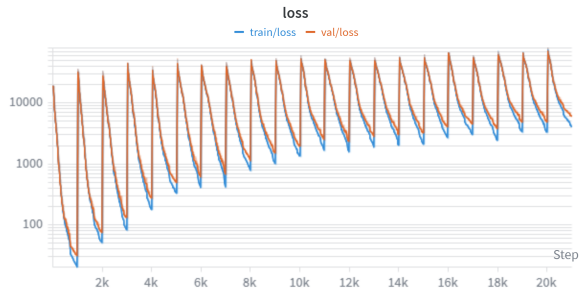
Latent Space: Spatial Truncation Error per Frequency

Loss increases monotonically with frequency (up to 17% at $k = 20$). Low frequencies ($k = 0$) feature smooth gradients; high frequencies ($k = 20$) show widely distributed, highly oscillatory patterns.



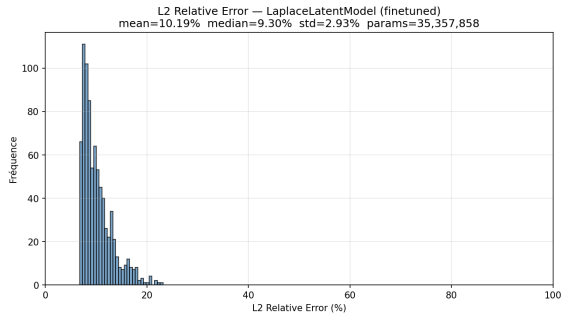
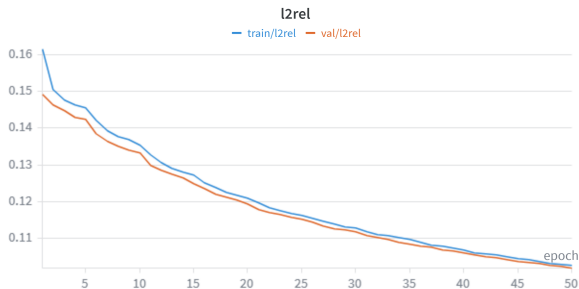
Phase 1: Frozen Latent Surrogates ($\theta \rightarrow z_k$)

21 Independent MLPs (one per frequency) directly regress parameter θ into precomputed z_k .
Observe the Sawtooth dynamics: High frequencies are increasingly difficult to predict.



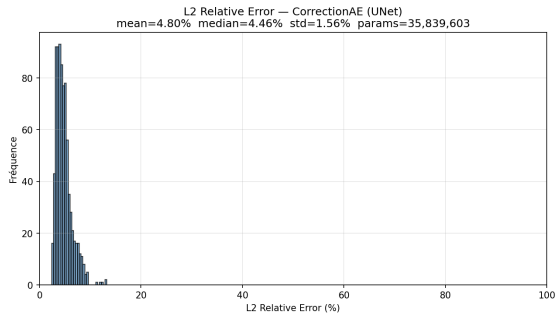
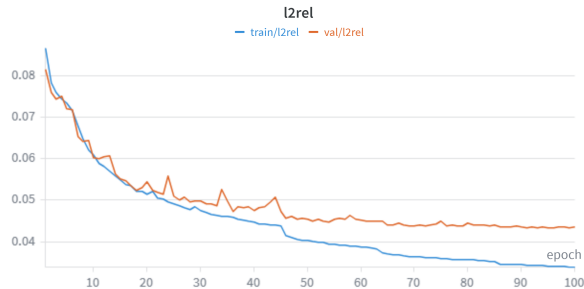
Phase 2: End-to-End Fine-Tuning

We unfreeze the AE Decoder and train the whole physical map end-to-end through the differentiable Laplace inversion. This directly minimises physical error, **halving the median error (9.30%)**.



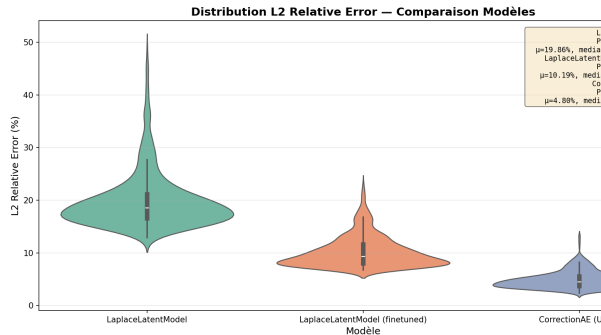
Phase 3: Residual Corrector (Gibbs Phenomenon)

Spectral truncation at $k_{\max} = 20$ creates inherent ripples. A 3D UNet (+481k params) suppresses ripples frame-by-frame on frozen surrogate predictions, reducing error to **4.46%**.



Complete Pipeline Benchmark Overview

Method	Median	Std	Range
Baseline	18.50%	5.35%	12.92 – 48.80%
Finetuned E2E	9.30%	2.93%	6.75 – 23.18%
+ UNet Corr.	4.46%	1.56%	2.37 – 13.28%



Conclusion

Key Takeaways:

- **Laplace Domain Decomposition:** Solves the memory bottleneck effectively, matching non-linear dynamics without losing spatial resolution.
- **Frequency-Conditioned Model:** Effectively builds a continuous non-linear space shared among complex frequencies.
- **End-to-End Fine-Tuning:** Absolutely indispensable for circumventing decoding bottlenecks (-50% error).
- **Post-Hoc Residual Correction:** An elegant, lightweight solution for eliminating spectral truncation footprints (-52% further error reduction).

Thank you!